S P E C T R A V I D E O Users Group (WGTN) ..

NEWSLETTER NUMBER 3 .... AUGUST 1984

The next meeting is on MONDAY , SEPTEMBER 17 , at 7.30 in the SEMINAR ROOMS 3,4,5 , Floor D , Wellington Clinical School.

We hope to have a review of the Bits and Bytes computer show , a short review of Ian Sinclairs book for those who have yet to see it , plus a question and hopefully answer time on Basic , CP/M etc and then demos and so on on the systems which will be set up . We can set up about 6 machines , so bring yours along .

The following meeting is on October 15 at the same venue and time

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Wellington SpectraVideo Users Group ....

Secretary .. Don Stanley
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Wellington Hospital
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Treasurer ... Colin Chin
P.O Box 2197
Wellington

At the meeting on Monday August 13 we made the following club decisions ....

Membership cards would be issued to club members (paid up subscribers) in order to obtain discounts being offered by both Moonshine Computers and Einstein Scientific. These will be posted to members with the next newsletter.

From 13/08/84 new members will receive newsletters from the one following payment of the subscription fee. Back copies of newsletters will be available while stocks last at a cost of $1.00 each

A5 size advertisements in the club newsletter will be charged for at a rate of $15.00 per month per advert

A club library is to be started , the object of this being to try and bring together as many reviews , books and other articles of interest to SV owners. Dennis Timmons is organising this , if you find an article of interest send a copy to Dennis , c/- P.O Box 7057 , Wellington

At club meetings dealers will be given a period of about 15 minutes to speak on any topic they wish to inform members of

Software written by club members will be sold by the club. For details contact Darryl Alison , ph 399510 during work hours.

The club will also sell any documents written by club members relating to SV .

BASIC NOTES ..... using the logical operations

In BASIC there are a number of operators (AND , OR , XOR , EQV , IMP , NOT) which carry out logical operations. There are also a number of operators (< , > , <> , <= , >=) which carry out arithmetic comparison operations. These notes are going to explore these operators in some detail and look at some not obvious extensions of some of them.

It is important to realise that the 6 logical operators are bitwise operators , they work on each bit of the operands separately. The first operator to look at is AND.

When an AND is carried out on two bits the following rule applies. An ON bit (binary 1) AND'ed with an ON bit gives another ON bit. Any other AND operation (ie one which involves an OFF bit (binary 0)) gives an OFF bit.

For example consider 15 AND 89

```
Binary 15   ... 00001111
Binary 89   ... 01011001
15 AND 89   ... 00001001   = 9
```

Note that any AND with a zero bit has given a zero bit in the AND. This is a very important tool in machine code where it is often necessary to reduce the first four bits to zero (mask them out). To do this you do an AND with 00001111 (15) which will force those bits to zero and leave the rest as they started.

Another useful application of AND is to force lower case input characters to uppercase. This program illustrates this ...

```
20 b$=chr$(asc(input$(1)) AND 95)
30 if asc(b$) > 64 and asc(b$) < 91 then goto 40
35 print "Invalid input -- Not between A and Z":goto20
40 etc
```

In this program b$ is the uppercase equivalent of whatever is input. Doing an AND with 95 leaves the input character as is if it is already uppercase , but makes it uppercase otherwise. A quick check that b$ is a letter is then carried out.

A further application of AND which is of relevance to the Spectravideo is with the SOUND statement in BASIC. This is covered in Appendix H of the 328 manual and in Sinclairs book.

Moving on to OR , the rule here is that anything OR'ed with a binary 1 gives a binary 1 , while 0 OR 0 gives 0. In other words , the result of an OR is OFF only if both bits are OFF.

```
      15  .. 00001111
      89  .. 01011001
15 OR 89  .. 01011111 = 95
```

Where do we need to use OR. Again it is of use in machine code , but has little use in most BASIC applications. However if you use the graphics PUT command you will find a use for OR (and AND and XOR). This command uses OR by doing a logical OR with the color of the pixel being PUT and the background colour (same for AND and XOR). Thus if you are using PUT on a yellow pixel (color 10) and use the OR option with a black background , the pixel will be PUT as color 10 OR 1 = 11.

XOR , EQV and IMP also have little use in most BASIC applications. Certainly there are specialised programs which may use them but in general they are of most use in machine code programming. XOR is also used with PUT , the resulting pixel color being in the above example 10 XOR 1 = 11 (XOR switches a bit ON if one or the other bits are ON , and OFF if both are ON or if both are OFF) .

The other operator is NOT , this is of little use in BASIC. It is a particularly simple operator , it merely changes the bits. Thus an OFF bit becomes ON , and an ON bit becomes OFF. Where is this of use in BASIC. It is well disguised within the PRESET option of PUT. Using the PRESET option with the above example with the original color 10 , to determine the result of the PUT you do a NOT with the original color , and then AND the result with &h00001111. WHY. Because NOT changes all the bits , and our number 10 is actually 00001010 in binary. Thus its NOT is 11110101 and doing an AND with 00001111 yields the required upper 4 bits , ie we mask out the first 4 bits. Thus the background color does not affect PRESET when used as a PUT option. (Note that the PSET option leaves the colors unaffected) .

You are probably familiar with another use of the logical operators , in IF statements. For instance IF A=1 or B=2 then.. , the = is an arithmetic comparison operator , others are <>,<,>,<=,>= . These can be combined with the logical operators to provide complex IF ... THEN statements. But they can also be used to take some of the tedium out of IF statements. For example

```
40 IF a=1 then B=2
50 IF a=2 then B=3
60 IF a=41 then B=7
70 IF a=341 then B=45
```

is equivalent to

```
40 B=-2*(a=1) -3*(a=2) -7*(a=41) -45*(a=341)
```

and you can combine the conditions in brackets with ORS ANDS XORS and so on to your hearts delight. This is called a boolean expression , the brackets tell BASIC that the condition inside returns a -1 if TRUE and a 0 if FALSE. Thus if A is 41 , B will be 7, as (a=41) will return a -1 which is multiplied by -7. Like the ON <Value> GOTO statement this can save a fair bit of memory in programs which have a lot of complex IF statements . In return for this saving a little more thought is required to set up the statement.

ODDS & ENDS ...

sent in by Garry Clark .

"I've discovered a simple way to put spokes within a full circle or part of a circle ... all of this within the CIRCLE statement.

```
CIRCLE (X,Y), RADIUS , COLOR , START , END
```

The start position of the CIRCLE is at 3 o'clock. the value of 0 or a blank is normally used. However by using any -ve number eg "-1" this will link the end statement position on the circumference with the centre of the circle .

```
eg CIRCLE (128,96),90,15,-1,4
```

Heres a simple program to create a wheel with the number of spokes designated. Try N=300 for interesting pattern effects. N=8 will draw an 8 piece equal share pie.

```
10 INPUT " NUMBER OF SPOKES ";N
20 CLS:SCREEN1,0:PI=3.1416
30 FOR A=0 TO 2*PI STEP 2*PI/N
40    CIRCLE (128,96),90,15,-1,A
50 NEXTA
60 GOTO60
```

Thanks Garry . (An interesting variation on this is to replace the (128,96) with STEP(A,A) ... ed)

If you're wanting a little more knowledge of SPECTRAVIDEO INTERNATIONAL then here goes ...

" .. SVI in Hong Kong has 4 factories and employs more than a thousand staff , its major shareholder being Bondwell and their registered offices being in the United States . One factory is devoted entirely to 328 production , one to software and the other 2 to 318 and peripherals development ..... "

Thanks to Computer Distributors for this information. And just to top it all off , a new portable micro has been released in New Zealand called .. the BONDWELL portable computer. So the major SV shareholder is now into the micro field in its own right. The BONDWELL will read or write OSBORNE , KAYPRO or SPECTRAVIDEO formatted disks , and comes with a bundle of software including Wordstar and other CP/M software . There are 2 5.25" disk drives built in , a 9" amber screen with 80 col capability , 16 function keys. There are 2 models , the main difference being that one has double sided disk drives , the other single , the double sided version (the model 14) also runs CP/M 3.0. Both systems have a built in speech sythesiser. If you want more info , contact CHECKPOINT COMPUTERS at TAWA or read the review in Computer Scene , August 1984.

Bits and Bytes magazine want to publish SpectraVideo programs in their magazine . They pay for programs , so if you have programs , even small simple ones contact Darryl Alison ph 399510 for more details.

SV reviews ... WHICH MICRO , MAY1984
COMPUTER CHOICE , MAY1984

Members ... our membership is growing rapidly ... new members include A.C Morrison , Chris Bossley , Lester Reilly , B.S Clark , Andrew & Richard Grenfell , Gavin Knight , W McCulloch , Andrew McMillan , Ross Sutherland , G Hughes , Darryl Alison , Jonathan Clark , Tim Miller , Terry Manning , Gordon Kindell , Ron De Bock ,K Markham , Geoff & Lynley Whitehouse , Graeme Neargy , Ray Martin , Einstein Scientific (P.North) , Justine Pickering , Harry Paterson , E R Dennis , Steve Hansen .

Members Discounts ... Moonshine Computers in Lower Hutt and Wellington as well as Einsteins Computers are offering discounts on selected goods to members .. enquire at the shops for details.

A mathematical function that seems to be omitted from most of the manuals is MOD . This fuction requires two arguements and returns the remainder after division. Thus X=15 MOD 10 will give X the value 5 as 15/10 = 1 and 5 remainder.

Saving space in programs ... if you have a variable which will use only INTEGER values you can save some program space by adding

A BASIC MONITOR.

This program lets you work directly in machine code. All addresses and data are entered in hexadecimal, e.g 8c (don't add &h at the front). Leading zeroes are not important.

There are 3 modes, selected by the special function keys 2,3,5.

"Address" (key2) displays the contents of a block of memory of any size.(no.of bytes entered as decimal)

"Data" (key3) lets you both examine memory and enter new data. You input an address and the contents are shown in ().

You have a choice of:

1. Just press ENTER to increment address.
2. Input data (then press ENTER) to change the contents of memory and increment address.
3. Input / to move to a different address.
4. Input z to escape from this mode.

" Run M/C" (key5) runs the m/c program starting at the address given.

```
10 REM  SV MONITOR
20 cls:key2,"Address";key3," Data":key5," RunM/C"
30 forI=2to5:key(I)on:next
40 ?"Press key 2,3 or 5 to continue"
50 on key gosub,70,160,,240
60 goto 60
70 cls:input"Start address, no. of bytes";A$,N
80 A=val("&H"+A$)
90 ?A$;
100 forI=1to8:p=peek(A)
110 ifP<16then ?" Ø"hex$(P);else ?"  "hex$(P)
120 A=A+1:N=N-1
130 ifN=0then return 40
140 next
150 A$=hex$(A):?: goto90
160 input "Address";A$: ifA$="z"then return 40
170 A=val("&H"+A$):?A$" (";:if peek(A)<16 then?"Ø";
180 ?hex$(peek(A))") ";
190 inputD$: ifD$="z" thenD$="": return 40
200 ifD$="/" thenD$="": goto 160
210 if len(D$)=0 then 230
220 D=val("&H"+D$): pokeA,D
230 A=A+1:A$=hex$(A): goto 170
240 input"Execution address";A$
250 X=0
260 A=val("&H"+A$): defusr=A
270 X=usr(X)
280 return 40
```

****

Try running this:

```
10 FORI=0 TO 255 : OUT130,I :NEXT
```

!!!

Thanks To GARTH Thornton

Spectra-Video Users Group (WGTN)

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a percent sign (%) to the end of the variable name. This is because the SV Basic automatically assumes all numeric variables are to be double precision and uses 8 bytes of memory to store them , using the integer declaration uses only 2 bytes.You can also use the DEFINT statement to define a lot of integer variables , you don't need the % sign then.

And from Garry Clark here is some more interesting graphics information ...

" Using SCREEN 2 will enable you use a PAINT statement which paints in a different colour to the border. There is an extra generally undocumented parameter for the PAINT command , namely the border colour. Suppose your border color is black (1) and you want to color inside green (2) . You use PAINT (x,y),2,1 . This command does not appear to work in SCREEN1 ... "

(however in SCREEN1 you could redo the command which created the border after painting , but using a different colour ..

eg .. CIRCLE (128,96),50,1:PAINT (128,96),1 and then
CIRCLE (128,96),50,13 .. will give a circle with
a border colour of magenta and painted black .. ed)

" further to the note on ARTIST in last months newsletter another problem is that a black background will lose the cursor which is fixed at black. The following modification will cure this problem ......

add the following new lines ...

```
  10 E=15
  99 gosub9550
9550 line (248,180)-(255,192),10,bf:locate249,182:print"+"
9560 if e=15 then e=1:return
9570 if e=1  then e=15:return
```

and change the following lines ...

```
15 .. add ,9550 to the end of the KEY statement
16 .. add :key(9) on  to the end of the key on line
310 .. change (x,y),1,1 to (x,y),e,1
1050 .. change  (128,96),1,1 to (128,96),e,1
```

Now to change the cursor colour you press function key 9 (F 9).

When you save these changes do so on a different cassette to the original.

Direct cursor addressing ... page 7 of the CP/M handbook describes the SV's ability to emulate a VT-52 terminal. Most of the escape sequences are fairly clear with the exception of direct cursor addressing . A bit of experimentation and access to a Digital terminal handbook reveal the following ...

" direct terminal addressing allows you specify the row and column positions to print at on the screen. The possible values for the lines are octal 40 to octal 67 , and for the columns octal 40 to octal 157. However you need to use the character , not the ASCII code to use this feature. To print at position 10,15 (LOCATE 10,15 in Basic) you need the following ....

```
10  a$=chr$(27)+"Y"
20  row$= chr$(&o40 +10)
30  column$= chr$(&o40 + 15)
40  print a$+row$+column$;"Hello"
```

This feature is of little use in the rom or disk basic as LOCATE is available , however in CP/M's MBASIC or BASIC compiler , and in machine code , this feature is necessary ."

Contributions to this newsletter are always welcome. Preferably type them on A4 with a good quality ribbon. Anything from software/hardware reviews through to notes on Basic , CP/M , machine code or whatever are acceptable . Send to Don Stanley.

Wanted ..... I have had requests for .. educational programs , in particular science and earth science related , and also chess programs . If anyone out there has written any please let me know or let Darryl Alison know if you want to sell them through the club . Also any ideas for a graphics screen dump program ??

MORE CORRECTIONS TO MANUALS ...

THE SV-902 DISK BASIC USERS GUIDE

PAGE	ERROR
8	4th to last line should read ... characters in length with an optional 3 character ...
10	the format for SAVE should be SAVE "1:filename"
16	line 70 of DEMO#4 should have GOTO 110 , not 120
29	in the description of LFILES it is stated that a cluster is half a track , this is WRONG , it is a whole track (implying that at most 39 files can be put on a 40 track disk with 1 directory track)

33	the DSKO$ statement does not include the <string expression> parameter. Read page 28 (top) for the correct way to use this statement

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**** FILE ALLOCATION TABLE ERRORS IN DISK BASIC ****

Never worried about disk backups , always careful about how you handle disks and then for no apparent reason when you type FILES back comes the reply BAD ALLOCATION TABLE and your disk is shot.

First the bad news , if it happens there seems to be no way around it . (If someone knows how to recover such a disk please let us know)

The good news .. how can there be good news with a problem like this ???

Several suggestions have been put forward to prevent the problem occuring . The first and most obvious is save every file on more than one disk ie make a backup. Another is never leave the disk on top of the drives , Never insert a disk while a file is open (a good rule is to type CLOSE before removing any disk) , you can get some weird and interesting disk problems by doing this. If possible make the first thing you do with any new disk a LOAD as this will input the new allocation table , unfortunately this is often impractical, especially if you are trying to save a program.

I am trying to find out from various people exactly what causes this problem , and if there is a cure for corrupted allocation tables .. keep your fingers crossed .

If you haven't found it already , SAVE "filename",s will save a graphics screen to cassette (1:filename to save to disk). On a disk a graphics screen save requires 4 tracks (it saves the whole VRAM)

A partial answer to the problem of retaining variables for a new program to use is provided by the FIELD statement in disk basic. The file buffer is not cleared when you load a program so you can store string variables in one program , load the next program and then recover the string variables. Restrictions are that you must have string variables , not numeric; and that only 128 bytes are used per FIELD statement.

FOR SALE ... Commodore 1701 colour monitor , 6 months use , video/sound leads converted to SpectraVideo , $700 ono . Contact Don Stanley